Bill of Materials:

Sl No.	Item	Price(in rupees)	Source
1	MIC-6 aluminium tooling plate blank, 200 × 200 × 30 mm (stress-relieved)	Approx 2700(850 per kg)	IndiaMart
2	Cartridge Heaters: ● 6.35 mm diameter, 230 V. ● Four 150 W × 125 mm, one 100 W × 125 mm, two 50 W × ~60 mm, four 25 W × ~20 mm. ● Stainless or Incoloy sheath rated 600 °C or more, with the diameter tolerance suited to a reamed hole.	Around 12000	Local Suppliers(for getting the exact specifications
3	PT1000 probes 6 mm, 3-wire, rated 450 °C or more	5000(1000 per piece)	IndiaMart
4	Toughened glass cookware lid, 24–28 cm	500	Amazon
5	Mica sheets(2 of 1 mm, each of 300*300mm)	1000(500 per piece)	IndiaMart
6	Ceramic fibre board, 25 mm	1000	IndiaMart
7	Thermal Paste	650	Amazon

Sl No.	Item	Price(in rupees)	Source
8	Sleeving(450 degree)	1000	Local supplier
9	2 ESP32 boards (1 spare)	600 (300 per piece)	Amazon
10	6 MAX31865 modules (5 plus 1 spare)	3000 (500 per piece)	Amazon
11	6 resistors, 4.3 kΩ 0.1%, for the PT1000 reference	300 (50 per piece)	Mouser / Digikey / Local supplier
12	6 branded 25 A zero-cross SSRs, plus 3 small heatsinks	4500	IndiaMart / Local supplier
13	5 V 2 A supply, 6 pull-down resistors, breakout board and headers	800	Amazon
14	Shielded high-temperature sensor cable	750	IndiaMart / Local supplier
15	Limit controller with K-type thermocouple	1500	IndiaMart / Local supplier
16	2-pole 16 A contactor	800	Amazon / Local supplier

Sl No.	Item	Price(in rupees)	Source
17	Latching E-stop	350	Amazon / IndiaMart
18	Double-pole master switch	250	Local supplier / Amazon
19	IEC inlet with fuse holder	150	Amazon / Local supplier
20	Line filter	450	Mouser / Local supplier
21	Fuses and holders: T6.3A, T4A, T2A, T1A × 2	300	Local supplier / Amazon
22	Terminal blocks, earth stud, cable glands	600	IndiaMart / Local supplier
23	Reference thermometer with K-type thermocouple	2500	Amazon / IndiaMart
24	3 dummy 100 mm wafers	4500 (1500 per piece)	Specialized suppliers / IndiaMart
25	RCD (optional, or use an RCD-protected socket)	1200	Amazon / Local supplier
26	Steel Box enclosure	1000	Amazon / Local supplier

Total Cost: 47,400 rupees

Work to be done

Workshop jobs

1. Machine the plate to 180 mm dia × 25 mm and grind the top flat to ±10 µm.
2. Drill the 7 rod holes (6.35 mm, about 127 mm deep, at the two depths), the 4 vertical trim holes from underneath, and the 5 sensor holes. The center sensor hole is about 90 mm deep and needs gun drilling.
3. Fabricate the stainless hot module, with a 186 mm cutout in the top panel, the base plate, and the heat shield.
4. Fabricate the aluminium or steel electronics bay with vent slots.
5. Cut the mica and ceramic board to size, with holes for leads and standoffs.

Electronics work

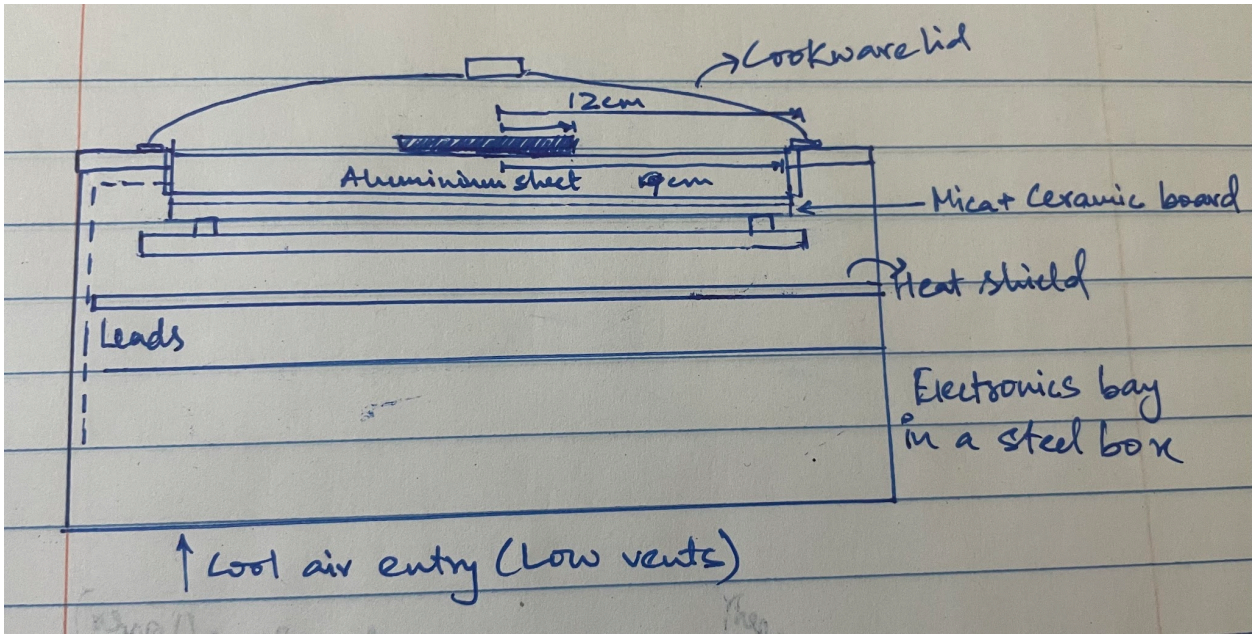
6. Replace the reference resistor on each MAX31865 module with 4.3 kΩ and set the 3-wire jumpers.
7. Wire the SPI bus and chip-select lines, and test each sensor against a known temperature.
8. Wire the SSRs, fuses, and 5 V supply on the bench with dummy loads first.

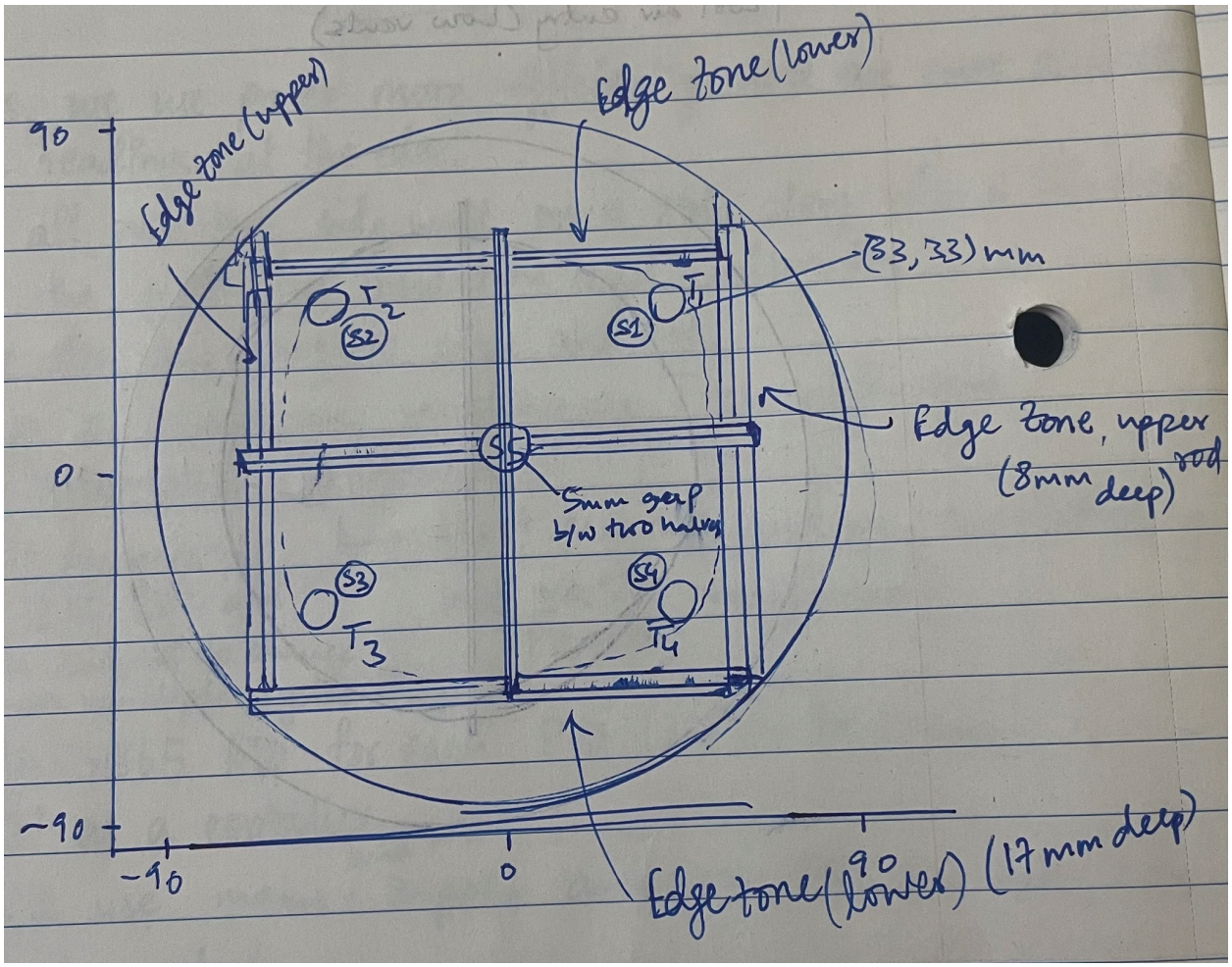
Assembly

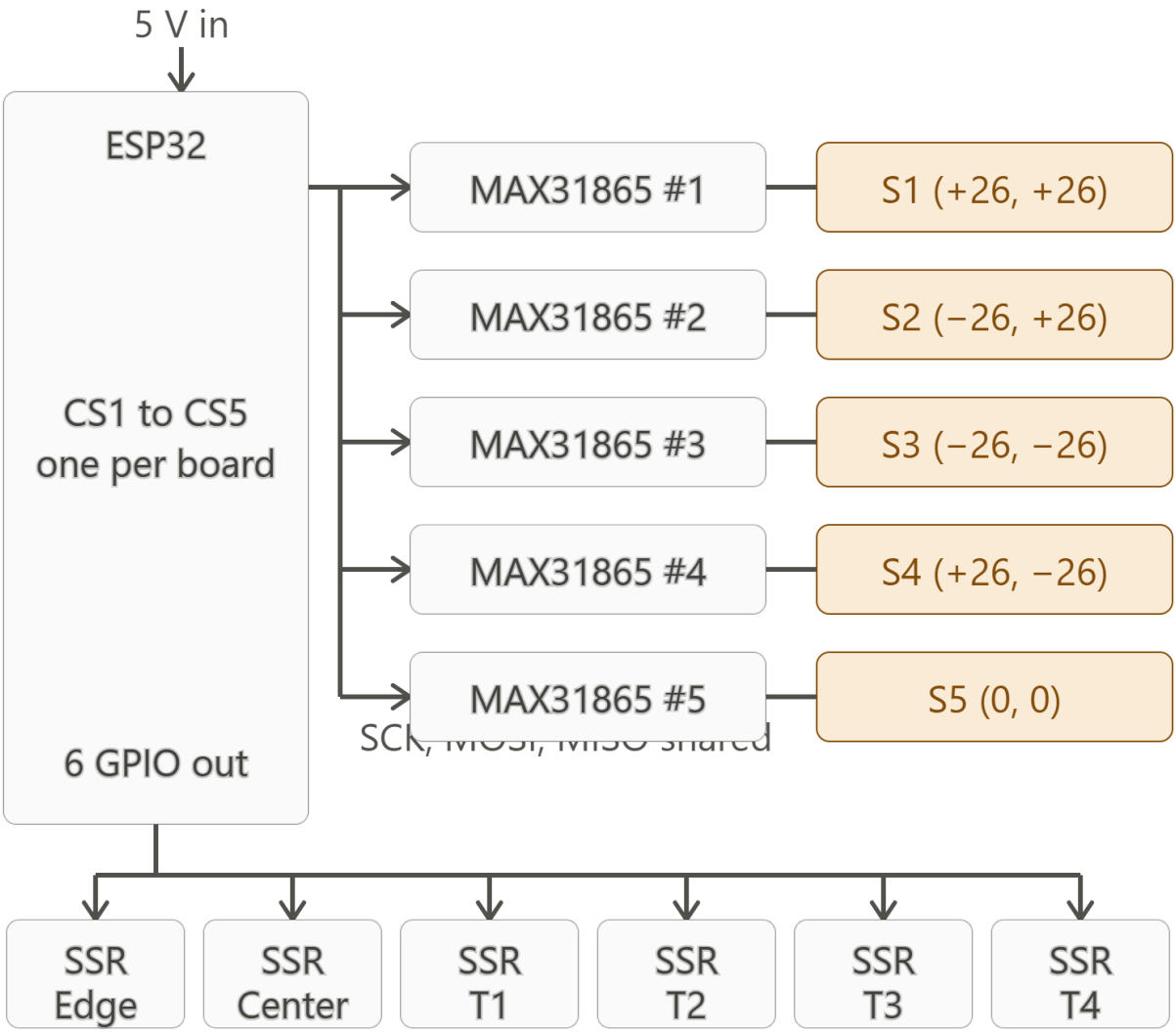
9. Insert the heaters and sensors with thermal paste.
10. Stack mica, ceramic board, and standoffs, and mount the plate in the module.
11. Fit the heat shield, route the leads through the ceramic-lined slots, and mount the dome seat pads.
12. Wire the mains section and bond everything to earth. Have a qualified person check it before first power-up.

Software and calibration

13. Flash the firmware, check the pin assignments, and tune at about 100 °C.
14. Calibrate each sensor against the reference thermometer.
15. Map the surface with the dummy wafer, set the edge offset, and tune toward 400 °C.


Cookware lid
12cm
Aluminium sheet
9cm
Mica + Ceramic board
Heat shield
Leads
Electronics bay in a steel box
Cool air entry (Low vents)


Edge zone (upper)
Edge zone (lower)
90
0
-90
-90
0
90
(33, 33) mm
S1
S2
S3
S4
S5
T1
T2
T3
T4
Edge zone, upper rod (8mm deep)
5mm gap b/w two halves
Edge zone (lower) (17mm deep)


5 V in
ESP32
CS1 to CS5
one per board
6 GPIO out
MAX31865 #1
MAX31865 #2
MAX31865 #3
MAX31865 #4
MAX31865 #5
S1 (+26, +26)
S2 (−26, +26)
S3 (−26, −26)
S4 (+26, −26)
S5 (0, 0)
SSR
Edge
SSR
Center
SSR
T1
SSR
T2
SSR
T3
SSR
T4